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Convergent encryption enabled secure data deduplication algorithm for cloud environment

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Summary

The exponential growth of data poses a critical challenge for cloud storage systems. Redundant data consumes valuable storage space and increases infrastructure costs. Data deduplication, a technique for eliminating duplicate data copies, offers a promising solution. However, existing deduplication techniques often compromise data security, especially when dealing with encrypted data. This paper proposes a novel approach that merges convergent encryption (CE) with data deduplication. CE leverages user data itself to generate unique encryption keys, enabling secure deduplication on encrypted data. We analyze existing literature on secure data deduplication and categorize various techniques using UML activity diagrams. We then present our proposed CE-based deduplication system, outlining its functionalities through UML diagrams. This research contributes to the field of secure data storage by proposing a novel and secure deduplication approach. By demonstrating its efficiency and security benefits, this work paves the way for more efficient and secure cloud storage solutions. Finally, we demonstrate the system's effectiveness through a comparative analysis, highlighting its potential to significantly improve storage efficiency while maintaining data security.

KEYWORDS

approaches, cloud computing, convergent encryption algorithm, deduplication, servers, UML diagrams

1 | INTRODUCTION

Web clients have significantly increased in recent times, with individuals and organizations increasingly adopting online modes of operation, which have become integral to their lifestyle and social interactions. The COVID-19 pandemic has further accelerated our reliance on the internet, with innovations enabling us to accomplish tasks like never before. Many sectors, industries, and educational professionals have benefited from technology to streamline various aspects of work. However, these advancements also expose us to significant cybersecurity risks, with online threats increasing sixfold. Attackers can exploit vulnerabilities to gain control of user systems and attempt to steal or manipulate data.

Cloud storage has seen a surge in user adoption due to its benefits in knowledge sharing and storage cost reduction. However, accessing vast amounts of data stored in the cloud presents challenges, including the need to sift through explicit data and manage resources efficiently to save data. Various techniques such as Proof of Ownership, Multiple Line Encryption (MLE), homomorphic encryption, and oblivious pseudo-random function are employed in the deduplication process.[1] These techniques are implemented at the block or file level, with additional protocols like secret sharing and the HMAC SHA algorithm used to increase storage capacity and reduce upload bandwidth. The convergent key management scheme is utilized to detect duplication in cloud storage, ensuring secure and reliable networking through distributed convergent keys.[2] While cryptography enhances data security in the cloud by encrypting data before storage and during retrieval, it alone cannot meet all data preservation needs.[3] Information leakage is possible when obtaining encrypted cloud data, necessitating research to maintain the confidentiality of encrypted data. The increasing

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use of computers, information, online applications, and mobile computing has led to exponential data growth, raising concerns about storage capacity. Cloud computing advancements have also led to concerns about vendor lock-in solutions and data security issues, despite the affordability of cloud-based services. Global statistics indicate a massive increase in data volume, with cloud capacity utilization visualized in Figure 1. Data deduplication is gaining attention as cloud service dependency grows, with the removal of duplicate data copies from servers optimizing storage efficiency.[4] However, combining encryption and deduplication is complex, as encryption aims to randomize content while deduplication identifies data similarities.[5] The use of a convergent key in encryption and decryption mechanisms facilitates secure key management and retrieval, although compromised keys pose security risks.[6] Addressing key management challenges is crucial, prompting the search for comprehensive solutions that ensure reliability, accessibility, confidentiality, integrity, and effective key management.[7] This paper proposes a novel approach that merges data deduplication with convergent encryption (CE). By leveraging data to generate unique encryption keys, CE enables secure deduplication on encrypted data. This approach offers enhanced data security by ensuring confidentiality throughout storage, improves storage efficiency by eliminating redundancy, and mitigates vendor lock-in by addressing security limitations associated with encrypted data deduplication. This research contributes to secure cloud storage by proposing a novel and secure deduplication approach for encrypted data, paving the way for secure and efficient cloud storage solutions.

The following are the article's primary contributions:

i. The analysis classifies data deduplication techniques, emphasizing their security implications with encrypted data and highlighting existing limitations.
ii. The paper proposes a new method merging convergent encryption (CE) with block-level deduplication, ensuring secure deduplication on encrypted data, addressing a key cloud storage challenge.
iii. The paper outlines the advantages of the CE-based deduplication system, emphasizing improved security and efficiency over traditional methods.

This research work draws upon a range of sources, including prestigious journal publications, legal cases, and data from diverse research institutions. The manuscript is structured into six sections. In Section 2, cloud computing-based data deduplication security is discussed. Section 3 provides insights into the background, evolution, methods, and benefits of deduplication. Moving forward, Section 4 delves into secure data deduplication approaches, offering a detailed exploration of their features and the latest proposals in this domain. In Section 5, the manuscript introduces

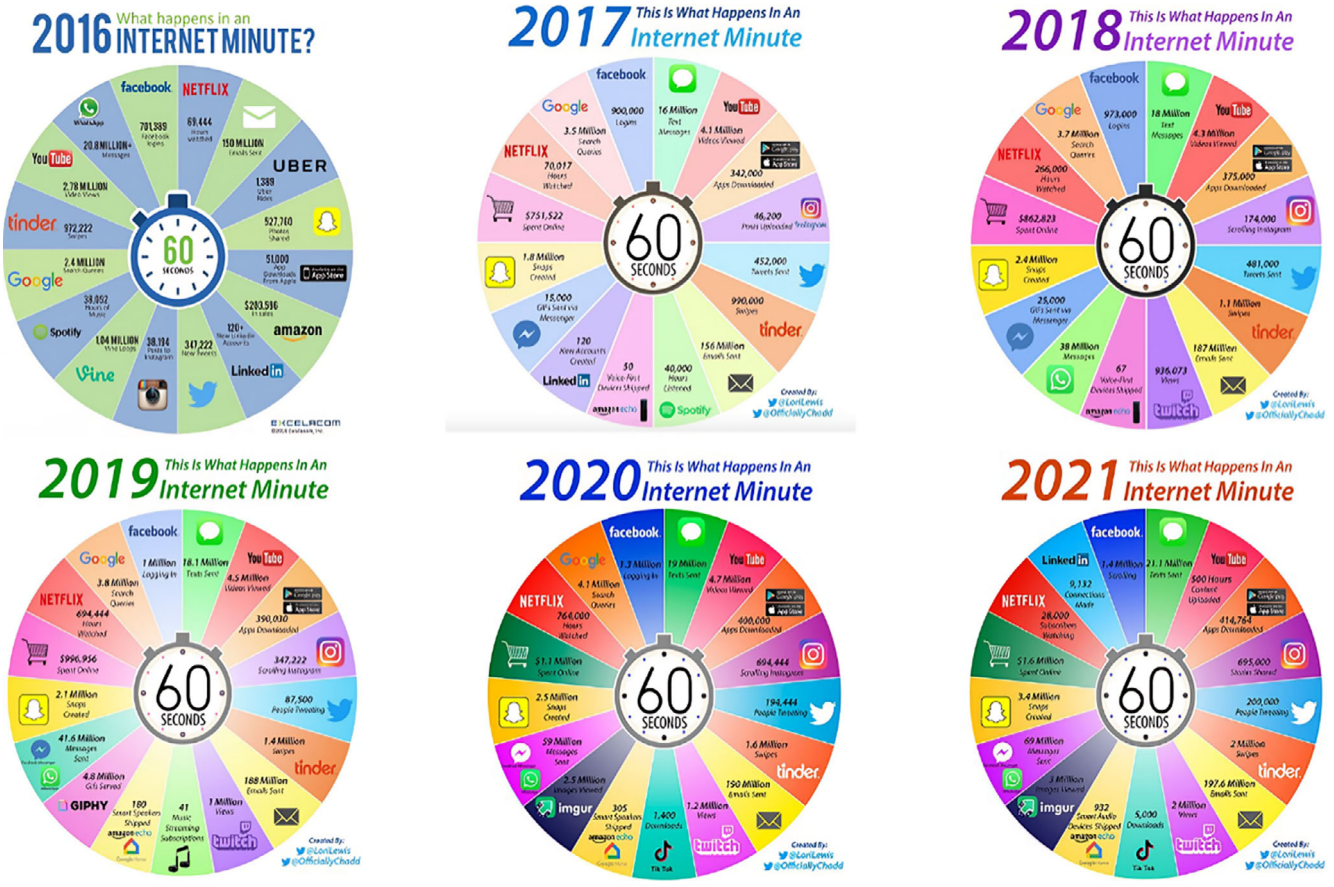


FIGURE 1 Infographic of a minute on the internet (2016–2021).

convergent encryption for secure data deduplication, accompanied by experimental analysis, security assessment, and comparisons with alternative methods. Finally, Section 6 serves as the conclusion, summarizing key findings and suggesting avenues for future research.

2 | CLOUD COMPUTING-BASED DATA DE-DUPLICATION SECURITY: A CONUNDRUM

Cloud Computing (CC) facilitates remote access to available resources by controlling and providing software and hardware resources. CC is an independent platform. There is no requirement to install any software on the Personal Computer (PC) to access the resources remotely. "CC is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction." This definition has been given by References 7,8. The use of the cloud and its growth in recent years has been a major benefit and convenience for society as cloud computing helps to lessen the need for resources for data storage. It is simple to access the data stored in the cloud from any location at any time. IT resources, including a physical server, virtual server, software, service, storage device, and network device, are all included in the cloud. The term "cloud service" refers to the software API used by remote clients (such as medical and paramedical professionals) to access resources.

There are merits to cloud computing for all types of end users. For instance, a company could share a sensitive file for business objectives or a regular person could save their information there. A de-duplication mechanism has been devised for significant data utilization in the cloud context.[9] The number of users of cloud services is increasing daily due to the advantages of storage and process models. Due to the inflated number of users, the amount of data stored in the Cloud will continue to increase, and a third party may be able to access the Cloud. Despite its benefits in terms of quick information exchange, lower storage costs, and need for more resources, it has security access issues. Finding a chunk of explicit information from the large pool is a new drawback of cloud services.

There are two primary drawbacks of cloud computing, that is, security and reliability. This is due to the ease with which any other client can access the user's data. For instance, hackers might try to access client data using valid usernames and passwords, changing the data, and so forth. Security in the cloud can be ensured via several methods, including encryption, authorization, and authentication. Both users and cloud service providers expose the cloud to security risks. Some risks related to cloud security include data loss, hacking, denial of service, malicious insider assaults, and issues with shared technology. Authentication, authorization, data protection, and other security considerations are just a few things cloud service providers need to consider. As data goes to the cloud, these fundamental security objectives are more important than ever. If a customer has confidence in the cloud service provider (CSP) and the services it provides, that confidence will likely play a significant role in whether they

TABLE 1 Data breaches in the years 2019 and 2020 with different attacks.

References	Records	Data breaches involved in various activities (%)
Cybersecurity Breach Report: Black Hat Ethical Hacking (2019)	12.3 billon	• "48% of businesses identified at least one attack per month." • "62% of businesses can respond to a breach immediately." • "67% of organizations are reported being breached at some of the points in the past systems." • "31% of organizations experienced cyber-attack on their operation infrastructure." • "80% attacks from phishing attacks." • "28% come from spoofed mail attacks." • "27% of attacks came from malware, ransomware, and spyware-related attacks."
Report on Verizon Data Breach Investigations (2020)	3950	• "70% involved in external actor's breaches." • "55% involved in organized criminal groups." • "30% involved internal actors." • "1% involved partner actors." • "1% featured multiple parties." • "72% of breaches involved large business victims." • "58% of the victim had personal data compromised." • "28% of breaches involved small business victims." • "86% of breaches involved in financially motivated." • "43% of breaches involved in web application." • "37% of breaches stole or used credentials." • "27% of malware incidents were ransomware." • "22% of breaches involved phishing."

decide to use a cloud platform or continue with the legacy framework. Trust is reliant on the CSP's obligation to protect data from all dangers, VM security, and other legal factors. During the cloud system security review (CIA), "Confidentiality, Integrity, and Availability" are the three factors that are considered.[10] The cost of each attack in 2019 was 3,83,365 US Dollars, as per the Cyber Security Breaches Report. It also states that 12.3 billion additional records were compromised in 2019. This is shown in Table 1.

Users can quickly move to meet their requirements by sharing a vast pool of shared assets offered by cloud data storage. To offer consumers storage services, cloud storage merges numerous storage devices using "application cluster," "web technology," and "distributed file system." Many cloud service providers exist in the literature like "Amazon Drive," "Apple iCloud," "BigMIND," "Certain Safe Digital Safety Deposit Box," "Cloud Me," "Crash Plan," "DriveBox," "Dropbox," "Google Drive," "IBackup," "IDrive," "Microsoft OneDrive," "Mega," "Mozy," "NextCloud," "pCloud," "SpiderOak," "SugarSync," "Sync.com," "Team Drive," "Ubuntu One," "Wuala" are a few of Cloud capacity suppliers expressed over offer cloud capacity with distinctive sorts of security like "Confidentiality," "Integrity" and "Availability." According to Reference 11, cloud security controls provided in different CSPs (Like AWS, Azure, Cloud, Oracle, IBM, and Alibaba) are presented in Table 2. Our observation from Table 2 is that if the objective is to achieve complete security controls, then Microsoft Azure should be prioritized. For instance, if any reputable organization's email protection is necessary, Google Cloud can be given a higher priority than Microsoft Azure. For managing services like "backup and recovery," "vulnerability assessment," "patch

TABLE 2 CSPs like AWS, Azure, Google Cloud, Oracle, IBM, and Alibaba offer a variety of security control services.

Control services involved in cloud service providers	AWS	Azure	Cloud	Oracle	IBM	Alibaba
MFA	YES	YES	YES	YES	YES	YES
centralized logging & auditing	YES	YES	YES	YES	YES	YES
Load balancer	YES	YES	YES	YES	YES	YES
LAN	YES	YES	YES	YES	YES	YES
VAN	YES	YES	YES	YES	YES	YES
VPN	YES	YES	YES	YES	YES	YES
Governance risk and compliance monitoring	YES	YES	YES	By 3rd Party	By 3rd Party	YES
Backup and recovery	YES	YES	YES	YES	YES	YES
Vulnerability assessment	YES	YES	YES	YES	YES	YES
Patch management	YES	YES	By 3rd Party	YES	By 3rd Party	By 3rd Party
Change management	YES	YES	By 3rd Party	By 3rd Party	By 3rd Party	YES
Firewall and ACLs	YES	YES	YES	YES	YES	YES
IDS/IPS	By 3rd Party	YES	By 3rd Party	By 3rd Party	By 3rd Party	YES
WAF	YES	YES	YES	YES	YES	YES
SIEM & log analytics	YES	YES	YES	YES	YES	YES
Antimalware	By 3rd Party	YES	By 3rd Party	By 3rd Party	By 3rd Party	YES
DLP	YES	YES	YES	By 3rd Party	By 3rd Party	YES
FIM	By 3rd Party	YES	By 3rd Party	By 3rd Party	By 3rd Party	By 3rd Party
Encryption at rest	YES	YES	YES	YES	YES	YES
Key management	YES	YES	YES	YES	YES	YES
DDoS protection	YES	YES	YES	YES	YES	YES
PAM	By 3rd Party	YES	By 3rd Party	By 3rd Party	By 3rd Party	By 3rd Party
Email protection	By 3rd Party	YES	YES	By 3rd Party	By 3rd Party	By 3rd Party
IAM	YES	YES	YES	YES	YES	YES
SSL decryption & reverse proxy	YES	YES	YES	By 3rd Party	YES	YES
Container security	YES	YES	YES	YES	YES	YES
End point protection	By 3rd Party	YES	By 3rd Party	By 3rd Party	By 3rd Party	YES
Certificate administration	YES	YES	By 3rd Party	By 3rd Party	YES	YES

management," and "change management" in the application, Microsoft Azure or AWS can be given higher importance. On the other hand, any of the CSPs mentioned above can be used if the requirement is to achieve only "firewall," "log analytics," "key management," "encryption at rest," "DDoS protection," "identity and access management," or "multi-factor authentication."

In Table 3, different evaluation schemes of data de-duplication present in literature are discussed in context with "Confidentiality," "Integrity," "Availability," "De-duplication type," and "De-duplication level." Various proposed techniques can work with existing CSCs. However, the CE algorithm is mainly used for securing data de-duplication over the cloud.

The evaluation of different data de-duplication systems concerning parameters like "key generation cost," "encryption/hash cost," and "ciphertext size," along with security analyses for resistance against different attacks like "online and offline brute force attack," "compromise of the key server," "leakage from hash," "collision resistance," and so forth are deliberated in Table 4 for reference. Different notations used in Table 4 are "H: Hash of Data," "Mul: Group Multiplication," "Exp: Group Exponentiation," "U: User," "KS: Key Server," "SE: Symmetric Encryption," "ASE: Asymmetric Encryption," "CuE: Customized Encryption," "Pair: Pairing," "|F|: File," "|B|: Block," "|GT|: Size of group elements," "N: Number of Key Server," "t: Number of key share server (KSS)," "DP: Data Privacy," "DI: Data Integrity," "CR: Collusion Resistance," "FA: Frequency Attack," "OfBF: Offline Brute Force Attack," "OnBF: Online Brute Force Attack," "CompKS: Compromise of KS," and "LeHT: Leakage from Hash or Tag."

TABLE 3 Data deduplication schemes comparison with different levels and security.

References	De-duplication type	De-duplication level	Confidentiality	Integrity	Availability
11	SS	BL	AES-128	Yes	Yes
12	SS	FL & BL	AES	Yes	Yes
13	CS & SS	FL	Elliptic Curve	Yes	Yes
14	SS	FL	AES	SHA-256	Yes
15	SS	FL & BL	Yes	SHA-1	Yes
16	CS	FL & BL	AES-256 with CBC	SHA-256	Yes
17	CS & SS	FL & BL	Yes	Yes	Yes
18	CS	BL	AES-256	SHA-256	Yes
19	CS	FL & BL	AES	SHA-1	Yes
20	SS	FL	AES	Yes	Yes
21	SS	FL & BL	No	MD5	Yes
22	SS	FL	AES	Yes	Yes
23	CS	FL	AES	SHA-256	Yes
24	CS	FL	Blowfish	SHA-256	Yes
25	CS	FL	No	SHA	Yes
26	SS	FL	Yes	Yes	Yes
27	SS	BL	Yes	SHA-256	Yes
28	CS	FL	AES-128	SHA-1	Yes
29	SS	FL	No	Yes	Yes
30	SS	FL & BL	Yes	MAC	Yes
31	SS	FL	AES-256	SHA-256	Yes
32	CS	FL	AES-256	SHA-256	Yes
33	CS	BL	Yes	HMAC	Yes
34	CS	FL	Yes	Yes	Yes
35	CS & SS	FL & BL	No	SHA-1	Yes
36	SS	BL	AES-128 & AES-256	SHA-128 & SHA-256	Yes
37	SS	BL	No	SHA-1	Yes
9	CS	FL	Yes	Yes	Yes

TABLE 4 Analysis of different data de-duplication schemes based on simulation.

References	Simulation details	KeyGen cost	Hash cost/encryption	Size of ciphertext	Size of tag	Security analysis
28	C	H	H + SE	|F|	|SHA|	DI and DP
21	C++ and C	-	H	|B| + |F|	|MD5|	-
33	NA	H	SE + H	|B|	|HMAC|	DI and DP, CompKS
38	NA	Mul + H + Exp	SE + H + Exp + Mul	|F|	|G|	DI and DP, CompKS
11	C	KS: H + Mul + Exp	SE + H + Pair + Mul + Exp	$\|G_T\| + \|G\| + \|B\|$	|G|	OnBF, CompKS, LeHT, DP & DI
37	C++	KS: H + Exp	U: ASE + H + Mul + Pair + Exp KS:Exp	$\|G_T\| + \|G\| + \|F\|$	|G|	OnBF, LeHT, DP & DI
26	Python	U: Mul + H + Exp	U: Mul + ASE + Exp + Pair	|F|	|SHA|	DI and DP
16	Solidity and C++	H	CuE	|F| + |B|	|SHA|	DI and DP
27	HSM	H	H + SE	|F| + |B|	|SHA|	OnBF, CompKS, DP & DI
18	C++	H	SE	|B|	|SHA|	FA & OfBF
19	C++	H	SE + CuE	|F| + |B|	|SHA|	FA, DP & DI
39	C	H + Mul + Exp	(2t + 3) SE•G + Exp + Mul + Pair	$3(t+3) \cdot \|G_T\| + \|F\|$	|G|	LeHT, DP, DI & CR
22	C	H + 2Mul + 6Exp	SE + 6Exp + 2Mul + Pair	$\|F\| + 2\|G\| + \|G_T\|$	|G|	OfBF, OnB, FLeHT & DI
37	C	-	H	|B|	|SHA|	-
40	Java	H + 6Mul + 6Exp	ASE + 6Exp + 6Mul + Pair	$2\|G\| + 2\|G_T$	$\|G_1\| + \|G_2\|$	CompKS, DP & DI
36	Solidity and Java	H + 2Mul + Exp	SE + 2Exp + Mul + Pair	$\|G_1\| + \|G_2\| + \|G_T\|$	|SHA|	DI and DP
35	NA	-	H	-	|SHA|	DI
30	C	H + 2Mul + 2Exp	SE + 2Exp + 2Mul	|F| + |B|	|MAC|	OfBF & DI

3 | DATA DE-DUPLICATION

Cloud computing and statistical deduplication are currently the most efficient and effective methods in use. The demand for digital information has increased, justifying the adoption of record de-duplication in a wide storage network and the cloud. The enormous amount of data being stored on storage systems emphasizes the importance of creating more straightforward methods to get rid of unwanted data. With the development of statistical de-duplication and its multiple ways, cloud computing can eliminate superfluous duplicates. De-duplication reduces bandwidth usage, storage costs, and energy consumption. Data de-duplication can be classified based on location and size: at "Server-Side (SS)" or "Client-Side (CS)" and based on size: "File Level (FL)," "Block Level (BL)," and "Byte Level (ByL)." This classification is shown in Figure 2. One method for performing de-duplication in an encrypted domain is to use a convergent encryption algorithm.

Differently formatted files (File 1, File 2, … , File 5) are depicted in Figure 3, which shows the storage of the heterogeneous files in the file storage database using secure data de-duplication schemes. Files (File 1 (A), File 2 (B), File 3 (C), File 4 (D), and File 5 (E)) conversion into vectors is shown in Figure 4. The comparison of data de-duplication between files is depicted in Figure 5.

Figure 6 outlines the fundamental idea of a multi-cloud atmosphere. Any record will be apportioned and examined for changes, to begin with. The information broadcast avoids all parts that are, as of now, put away within the framework by any client. Henceforth, by using the approach of secured data de-duplication, where modern substances will be transferred and put away securely.

A flowchart of the concealing computation is shown in more detail in Figure 7. The flowchart visually outlines the step-by-step algorithm for determining how to conceal information. It provides a clear and structured representation of the decision-making process involved in concealing sensitive data or details. The process begins when a request for record transfer is received as part of a partitioned request. Initially, it verifies the presence of the segment by searching for its unique identifier in the database. If the segment is not found, a new storage reference will be generated randomly. The renaming operation to a new random title will be carried out after the upload is completed, and this step is necessary to obfuscate the subsequent replacement operation in the algorithm.

Data de-duplication, a crucial step in removing duplicate copies of information, is a component of data compression techniques frequently used in cloud storage to reduce storage capacity and implement data-saving procedures. Specific techniques are utilized on different forms of statistics, such as material, pictures, and videos, using distinctive de-duplication procedures. Each of the three record types has unique characteristics and excellent capacity positions. De-duplication systems include explicit methods to locate and delay copy records depending on the data.[41,42]

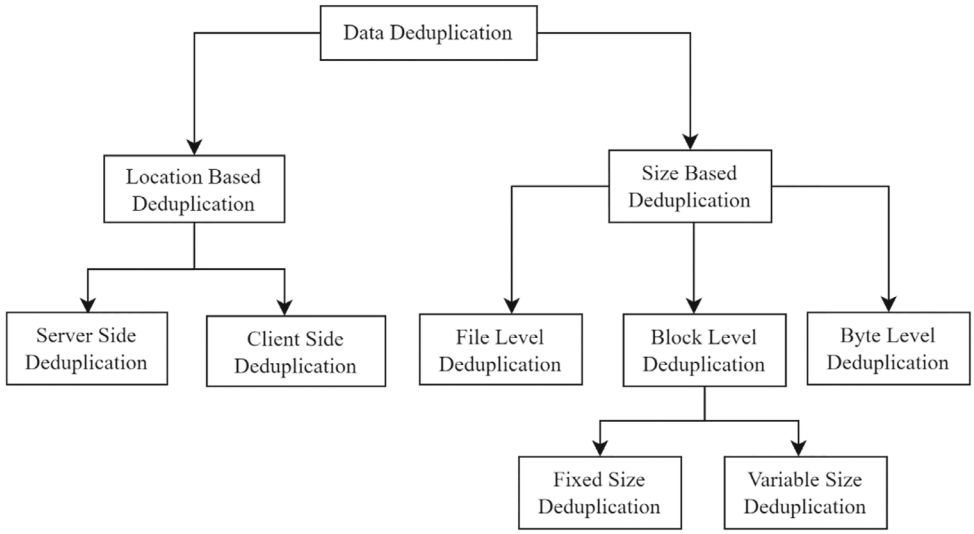


FIGURE 2 Classification of data deduplication.

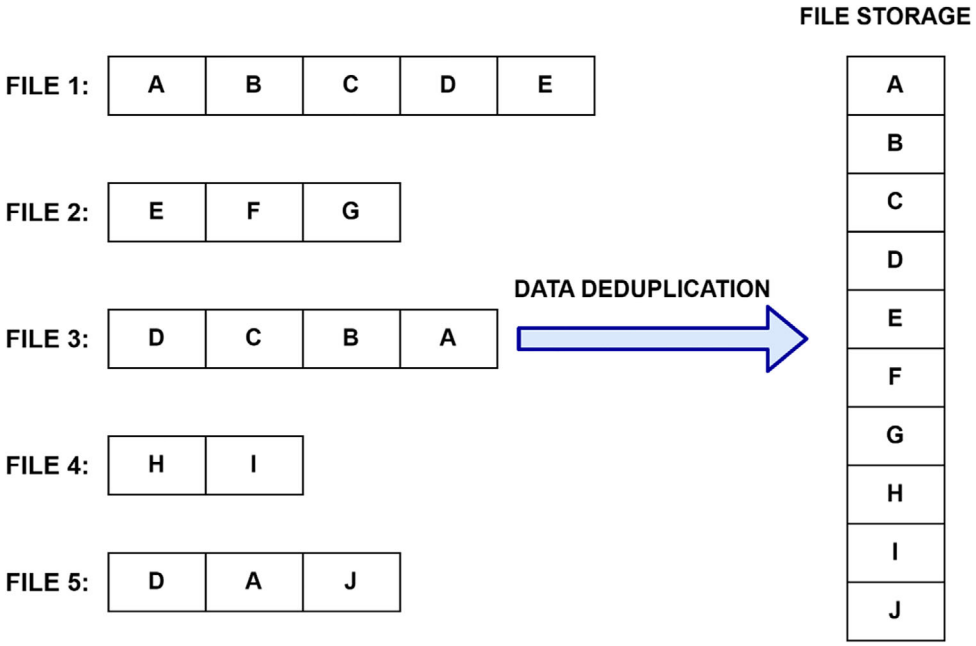


FIGURE 3 Data de-duplication for file storage.

Accordingly, the type of information is vital for enhancing the de-duplication processes. The records arrangement is essential for evaluating, getting, and organizing the points. It takes bit-stage blending to find duplicates in usable records. Because of the diverse configurations of statistics, there are different ways to examine copies in text, picture, and video. A small number of data duplicates, or "replication aspect," are kept in a sizable, distributed storage device to achieve high data accessibility. Any reproduction stats above the replication factor are evacuated to reduce the need for a garage, storage cost, calculation, and electricity. De-duplication techniques for large-scale distributed storage architectures have gained popularity in both the academic and commercial worlds due to their numerous benefits to the business.

However, these solutions need to be improved due to the effectiveness and sufficiency of ways to coordinate the facts. Researchers and business analysts are working to develop more practical distributed de-duplication techniques. The entire document is divided into chunks of fixed or varying sizes. According to Reference 43, a de-duplication system saves a single copy of each segment and uses rules for multiplication portions. Suppose that the de-duplication engine of the capacity device learns that this is stored somewhere inside. In that scenario, it stores a pointer in the duplicate information area that points back to the genuine duplicate. It makes it possible for the barriers inside the garage device to be removed, freeing up memory space.[44,45]

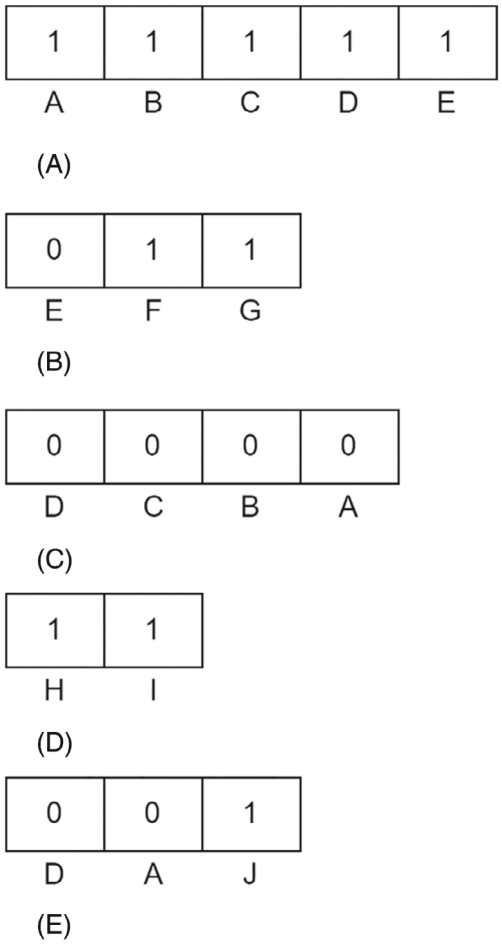


FIGURE 4 Conversions into vectors: Files 1 (A), 2 (B), 3 (C), 4 (D), and File 5 (E).

	A	B	C	D	E	F	G	H	I	J
A	1	0	0	0	0	0	0	0	0	0
B		1	0	0	0	0	0	0	0	0
C			1	0	0	0	0	0	0	0
D				1	0	0	0	0	0	0
E					1	0	0	0	0	0
F						1	0.3	0.2	0.1	0
G							1	0.6	0.5	0.4
H								1	0.8	0.7
I									1	0.9
J										1

FIGURE 5 Data deduplication comparison between files.

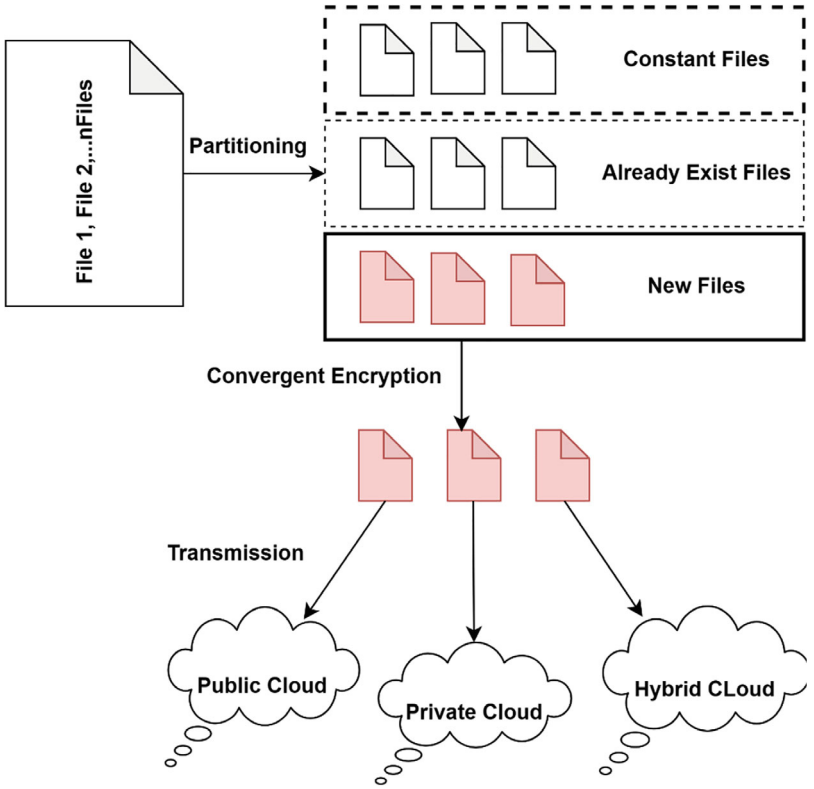


FIGURE 6 Secured cloud data deduplication overview.

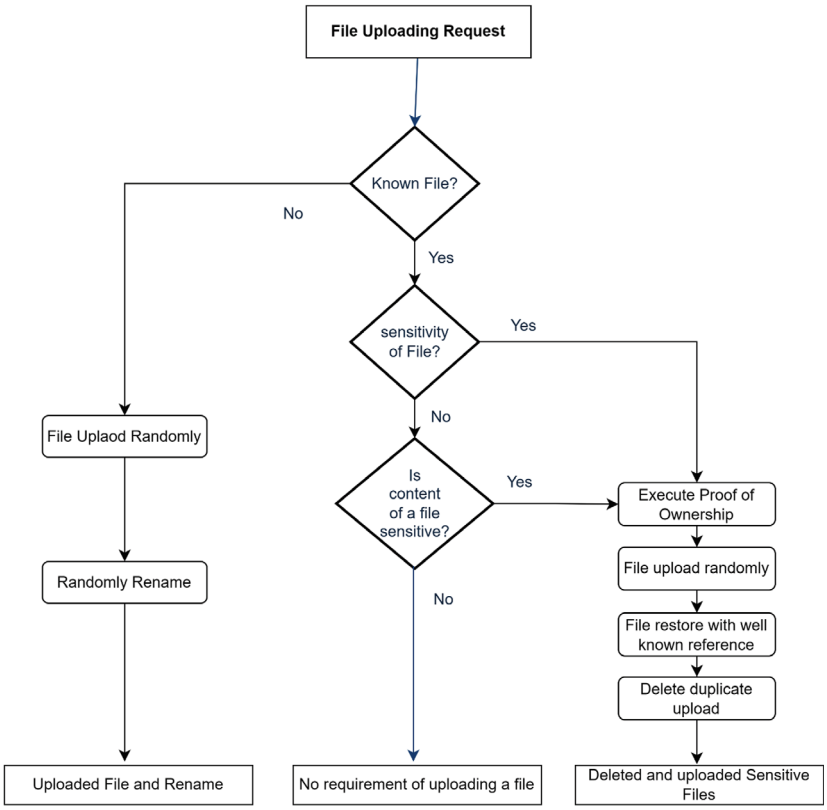


FIGURE 7 Flowchart of DAIC (decision algorithm of information concealment).

After reviewing the most recent research on de-duplication techniques, it was determined that it was necessary to carefully evaluate the text that was available on the subject. In any case, this stage summarizes the motivations, dedication, and interest in this subject.

i. The functions, necessities, benefits, and drawbacks of a de-duplication approach to improving the overall implementation of an extensive memory system have been discussed.
ii. The garage, point of utility and stage have all been considered when organizing the present de-duplication techniques. The de-duplication strategy has also been described as dependent on videocassette, photo, and literary content. It has been made clear how important content de-duplication techniques are to consider. There are numerous scientific classifications, picture- and video-based de-duplication, and content evaluation offered. As a result, this literature assesses writing and provides a broad overview of de-duplication techniques.[46]
iii. Future de-duplication research criteria were highlighted for academic and business experts.

De-duplication of facts is a phrase that was developed in the early 2000s to handle massive storage systems with high granularity.[46–51] It opposes conventional data compression methods that ignore repetition over a limited record organization that is heavily dependent on excessive intra-document information. Instead, the cryptographic hashes of each text or mass are used to figure out how to identify the copies. On mixed media substance, these tactics were also revised in 2008. By evaluating the degree of similarity between different photo or video frames, the highlight extraction and hashing techniques analyze the replicated multimedia content. These techniques for reducing record duplication emerged to address the issue of creating data length within storage structures proposed by References 52–54.

Bytes and strings can be compressed using Huffman and dictionary coding, while de-duplication algorithms eliminate log or bite redundancy. In addition, by using delta and loss-related compression techniques, redundant fact-reduction techniques dating back to the 1950s started to appear more frequently in the 1990s. Finally, in 2000, optical de-duplication techniques and computer de-duplication tactics arrived here.

Modern CRM and decision-assistance programs draw heavily on statistics warehouses, which are repositories of data collected from various statistics assets (Client Dating Control). Because choice-guided evaluations of data warehouses influence key business decisions, accuracy is essential. Facts sources can be used to observe impartial and almost incompatible standards since they are impartial. Most of the time, confusion affects group sites with diverse items with a similar percentage of nomenclature. In the finest recovery tool, a user can enter an entity or concept call and look for results grouped according to the unique entities/standards that percentage that call. Including more data in the listed files is one way to enhance such devices.[55] Businesses typically become aware of rational-specific discrepancies or inconsistencies while compiling data from several sources to create an information warehouse. Such issues fall under the category of fact heterogeneity. When data from numerous statistical sources are used to keep the information overlapping, the facts are foolishly repeated. However, there are grammatical flaws, conflicting statistical source conventions, omitted fields, and other issues with the data collected for the facts warehouse from external assets.

To provide for the provision of excessive statistical quality, arriving statistics tuples outdoor assets need justification and modification. High-quality statistics show that the data warehouse should operate "error-free" Techniques for record cleansing are essential for enhancing the first class of records. Information mining approaches successfully mined data using software or algorithms to produce artistic and useful analysis. Descriptive and extrapolative facts about fashion are two different categories. Illustrative representations like "clustering," "summarization," "affiliation Rule," "series discovery," and many others locate the homes of the investigated data using tracing patterns or association records. A group of substances is grouped by clustering at the number of subsets known as clusters, where items from the same cluster are relatively like one another and those from different clusters are dissimilar.[17,56–59] To consolidate and summarize the data, clustering is a fundamental technique that may provide a summary of the preserved information.[43] Predictive versions like classification, regression, time series analysis, prediction, and so on select whether to employ known values for an unknown information type technique. To build a version, the classification rules study the training set. The k-nearest neighbor approach (k-NN), the Naive Bayes algorithm, the neural network,[34,60,61] and the auxiliary vector mechanism are examples of strict classification techniques (SVM). The widely used type algorithm, k-NN, exhibits favorable performance uniqueness and is employed in many diverse applications, including text primarily dependent on image recovery data and three-dimensional item interpretation (evaluation of entropies and deviations). The process of "data cleaning," also known as "records cleaning" or "scrubbing," raises the caliber of statistics by identifying and removing data mistakes and anomalies.[62] By removing data set modification and reducing fact replication, it inspirationally improves the overall statistics compatibility. Modern information-removing techniques identify record replicas, unrecognized values, record and field resemblances, and replica deletion. The duplicate document detection technique is used to identify additional or multiple characteristics of one exceptional actual global object or item.

3.1 | Repetitive data minimizing methods

Repeated statistics reduction approaches have been created to control the expanding number of virtual records and select surplus at the byte, string, and report levels. Similarities exist between the business and its development of redundant statistics reduction approaches. A bit-price discount strategy is used to compress information to express data in a condensed manner. This looks for redundant information and reduces the amount of storage space needed. The lossless compression process serves as the standard definition of statistics compression.[60,61] The distinct, unique

information is rebuilt from the compacted data using the lossless compression technique. By highlighting useless data, such as the compression of JPEG photos, lossy compression reduced the details. The first-act prediction is reenacted in this. With lossy compression algorithms, data in movies and music is compressed.[41] This section contains essential historical background on redundant information reduction methods, demonstrating the development of each conventional method for lossless data compression, delta compression techniques, and information de-duplication techniques. Additionally, it provides a scientific classification of many approaches and their expansion.

3.1.1 | Lossless data compression methods

The term "information compression" was established due to its wide usage. Word reference encoding, run duration encoding, and entropy encoding are three techniques for data compression that do not sacrifice quality.[63] A few sequences in a little served as symbols for the character series. Vast amounts of superfluous statistics are generated and ejected in these strings as designs of facts.

Byte stage: Entropy encoding is used in the early record compression techniques to detect byte-level redundancy. Two entropy encoding methods are used to represent frequently occurring samples with fewer bits each: Huffman encoding and mathematical coding. The best prefix code was developed by D. A. ***Huffman using binary trees with frequency considerations. Variable-length codes are used in place of constant-length codes.[64] Short coding is used to present frequently used images. Elias created arithmetic coding in 1960 by 62 et al. (1987). The entire message is stored as a set of floating-point numbers within a predetermined range.

String level: In a different study, the String Level approach developed by Reference 65 was designed to review and eliminate repeated strings.

3.1.2 | Mechanisms for the compression of the delta

The delta compression method was developed in the 1990s to compress similar files or chunks. The two most popular applications are backup storage and remote synchronization. It searches similar chunks for matched texts using a sliding window that is byte-sensitive. Variations between sequential files and complete records are kept by Reference 55 in the "delts" or "diffs" form (2016). One of the elements of the Delta Compression Strategies is the string step. The delta computer uses a byte-clever sliding window during the X delta and Z delta string phases of delta compression to find redundant strings between the destination chunk and the source chunk.

3.1.3 | De-duplication methods for data

- *The records de-duplication strategy* was first implemented in 2000 to help with coarse-grained global compression. While statistical de-duplication techniques can be applied at the reporting or sub-reporting level, the methods outlined above need extremely long processing times and are not feasible. By using pieces of fixed or variable size, it compresses statistics. Using cryptographic hash functions, some chunks are assigned hash values, and duplicates are recognized by hash values that are the same. For record-level de-duplication at the file level, both methods are applied, and reporting is handled separately. This verifies the backup document's index to verify the attributes kept there. If an identical record is available, either an indicator of the current document is given, or the index price is updated and stored. The single instance garage, where it is simple to use the whole document hashing approach, is the best example of the study and has thus been saved because creating and processing report hash numbers is clean and uses very little energy. However, the paper contains a few one-byte exchanges that lead to novel technology that requires exclusive data. With the development of block de-duplication algorithms, the issue of document-level de-duplication is resolved.
- *Block-degree (sub-report-level) de-duplication*: These methods include breaking up a file into several compact blocks of either fixed or variable lengths. Comparable blocks are recognized using hash techniques like "MD5," "SHA-I," "Rabin fingerprinting," and "similar hash algorithms." As a result, a block is updated in its index and, unlike other blocks, gets written to the disc. In every other situation, the block of equivalent truth will be moved back to its original position. As a result, IDs will significantly grow, necessitating more processing power. Constant-duration or variable-length de-duplication and block-stage de-duplication are both types of de-duplication.[21] When analyzing fixed-length computer blocks utilizing de-duplication of the fixed-length block period approaches, the same statistics block is not duplicated. The disadvantage of the fixed-length block technique is overcome by variable-length innovation. There are blocks of statistics with different lengths. One kind of method is used by variable block algorithms to choose the block time. As a result, their chord block boundaries may "drift" within. To keep the borders of several block locations from altering when changes are made to one part of the structure.[62] The phase includes bytes in the duration in a content-based and restricted manner. It increases the granularity and flexibility of a block's management.

3.1.4 | Merits and demerits of data de-duplication

The subsequent deserves of de-duplication are identified as follows:

- *Lesser garbage area*: The amount of storage space needed to store archives, files, or various forms of data is reduced via de-duplication. The most precise reproduction of the statistics is stored, and duplicate copies are removed. Therefore, storing more data results in greater free space.[41]
- *Boost of community bandwidth*: It makes sense to recommend data duplication to avoid sending replication copies over the network as the precise copies are archived on the disc. That is, the amount of bandwidth needed by the community can be decreased through de-duplication.
- *Energy reduction consumed*: By lowering the demands for capacity and electricity, de-duplication is a capacity optimization strategy that minimizes energy usage. In decreased storage, less electricity and coolants are required. It reduces the burden on the tech gear and conserves energy.
- *Lessen usual garage value*: Significant time, space, network bandwidth, human resources, and financial savings are made possible through de-duplication. Additionally, it improves the effectiveness and efficiency of storage systems.

3.1.5 | Demerits of de-duplication

- *Impact on garage efficiency*: The main garage machine's fixed-size technique stores many chunks at different memory locations. It causes fragmentation issues that hurt the outcomes. For its execution, the de-duplication process needs additional resources like memory, recollection, and bandwidth. Any ineffective de-duplication strategy reduces the effectiveness of a sizable storage network.
- *Information integrity loss*: The data blocks are listed for easier hash value searching. Due to a hash collision, equal hashes can be generated for unusual data blocks, which could jeopardize the records' accuracy. Therefore, hash collisions must be managed correctly to prevent any data loss and keep the data's integrity.
- *Issues with backup devices*: Information relocation and technique may require a new hardware tool for fact de-duplication. According to Reference 66 such support equipment may also increase costs and impact storage performance.
- *Respect for privacy and security*: The de-duplication techniques can always be finished.
- *(Storage) Archiving*: It can be used from the repository's entrance. To prevent such security lapses and the loss of sensitive data on devices, the security of the deductibility mechanisms must be properly planned.

3.1.6 | The ANN technique

A synthetic neural network is a device that, in other words, mimics organic brain devices and is primarily focused on the operation of biological neural networks. A tool for specific approaches like categorizing, optimizing, and so forth, is an ANN. A neural network can perform tasks that a linear program cannot complete. The utilization of the neural community's parallel nature can continue without issue, even if a specific component malfunctions. A learning approach is implemented using a neural network, eliminating the need for further programming. Particularly, a synthetic neuronal culture can take the following forms:

- If data from input to output processors must be intentionally sent forward, the neural network is referred to as a feed-forward neural network. For example, records processing may involve several (layers of) devices. Still, no remarks are present—meaning that links running from device outputs to devise inputs in the same layer of lower levels.
- Recurring neural networks contain references to statements. Contrary to feed-ahead networks, the variety of residences within the community is crucial. In other cases, the activation levels of the devices undergo a resting process so that the neuronal population can form a stable nation where such activations no longer alternate. For the complex activity to form the neural network output, additional packages depend on changing the output neurons' activation values. Therefore, a multi-layered neural network was employed. The facts, hidden, and output layers make up the skeleton of a neural network. The following discernment may show the framework that governs the basic shape of a multi-layered neuronal community.

3.2 | Statistics of data de-duplication

These documents are duplicated on distributed storage devices for high dependability, accessibility, and disaster recovery as data in cloud storage expands tremendously.[42] For the device to be protected against errors and to maintain high availability, the replication factor, or minimum range of data replications, is essential. The replication component must be the only wide variety left in the garage system. In all other cases, duplicate data on the home computer adds to the strain due to the limited bandwidth and additional space. De-duplication techniques are used to increase the storage device's capacity for cost and use terms to lessen or manage this information duplication. The usefulness of de-duplication techniques depends on the nature of the facts, such as whether they are based, unstructured, or semi-structured. Statistics fit into the same categories as text, pictures, and videos. The maximum possible network traffic processing speed, overall storage capacity, and storage device efficiency are all impacted by replication

knowledge.[67] Researchers could comprehend how cutting-edge de-duplication methods for garage buildings are being built as a result. This involves sending green information to a garage unit and eliminating duplicate records. De-duplication is a technique for automatically removing reproduction statistics from garage systems.

With Microsoft's assistance, the de-duplication reduction of facts is reported. Over 4 weeks, 857 desktop Windows machines from NetApp Microsoft assessed the stability of space savings between full-document and sub-record de-duplication.[21] According to observation, block-level de-duplication only meets 32% of the initial requirements, whereas total record de-duplication has a gap reserve of 75%. Records de-duplication was also employed in a virtual library that used duplicate bibliographic information reports to detect the use of similar functions on two real datasets.[68,69] Datasets include article quotation data for the core collection and metadata information for two free virtual libraries (BDB Comp and DBLP). Think about how the high-quality metadata de-duplication enhances the digital library data set by 2%–62% and by 7%–188% in the item dataset. According to NetApp, de-duplication may remove 95% of replicate information inside storage structures.[51] According to experimental data, 95% of regular backup costs are covered by VMware, 30% by email, 35% by records, and 72% by regular backups.

3.2.1 | Classification of de-duplication *techniques*

Based on local and global deductions, the categories of storage, such as primary or secondary de-duplication, source/goal, and handling time, were divided into online and post-process deductions. The four criteria for de-duplication classification taxonomy are length, form, timing, and degree.

They are classified as follows[62]:

- *Depending on the storage type, de-duplication*

Based on the type of storage, de-duplication classification was carried out.[51–53] and Chandrasekar apply de-duplication for primary and secondary storage, respectively (2012). Initially storing: The main memory or active storage directly available to the CPU is where the primary storage-based de-duplication is executed. An additional or external storage device without direct access to the CPU is referred to as secondary storage. It backs up data from primary storage. Only historical data is stored in and retrieved from these systems. Storage archives, snapshots, and backup storage are a few examples.

- *De-duplication based on the type*

The supply side or the appropriate direction is used to carry out the de-duplication process. De-duplication is based largely on source and target, depending on these types. Before being transmitted to the backup objective, the statistics on the supply side are plicated.[21] The program remembers moving the data to the backup server and is installed on the server's CPU. Therefore, it also lowers the bandwidth, storage, and time requirements for information backup. On a dedicated garage device on backup servers, de-duplication is typically carried out on the duplicates. In this sense, all de-duplication functionalities—garage utilization with the added benefit of committed use—are addressed by dependable hardware de-duplication machines. Therefore, the statistics for complex garage systems may not have any overhead. However, as discussed later, it requires more resources and is better categorized as a publishing method.

- *De-duplication of data based on timing*

Timing-based de-duplication only applies at the instant the de-duplication algorithm is running. After that, it sets a deadline for completing the de-duplication process. De-duplication techniques like duplicate searching are the main remedy for timing-based de-duplication. It can be done using either an asynchronous/in-band operation or an asynchronous/out-of-band operation. The timing-based de-duplication was also divided into inline and post-system categories. Before being written to disc or on the source side, the deduplicated data is processed. Therefore, additional disc space is not required to maintain and defend the facts. Since the information is exceeded and processed most effectively when Inline de-duplication requires further processing, it boosts efficiency. In addition to being faster than inline de-duplication, it also goes by offline de-duplication.[47] This enables the backup time to be cut down.

- *Data deduplication based on level*

There are two types of data de-duplication: local level-based and global level-based. Only one VM can do local de-duplication, and only one node can identify replicas. Because it cannot eliminate all duplicates, it impacts overall performance.[42] It has a few facts and nodes. De-duplication, or noneducation of the report, is a process that is carried out in a distributed setting or across many datasets.

- *De-duplication of data in the cloud for storage systems*

The data de-duplication approach is frequently employed in cloud storage, backup environments, and data storage systems since it lowers the need for storage space and storage costs. De-duplication techniques use only one body replica to save data, which consumes more internet

bandwidth than the total amount of data sent to the cloud or the community. It promotes the acceleration of cloud backup,[70] leading to faster and more environmentally friendly information security operations.

Direct cloud de-duplication, secondary garage copies, and cloud gateway de-duplication can all be deployed for cloud storage de-duplication.[23] De-duplication can also be employed in unique garage systems, such as primary, secondary, and cloud storage platforms. Devices for personal, public, and mixed cloud storage all profit from the de-duplication technique.

4 | SECURE DATA DE-DUPLICATION: STATE-OF-THE-ART

In the past, numerous scholars have proposed several studies in data de-duplication for cloud storage. One of them was a cloudy system presented by Reference 27, which ensured block-level data de-duplication and confidentiality. It is carried out by removing redundant copies of data to limit cloud providers' capacity. De-benefits duplication comes at a cost, however, in the form of increased security and assurance risks. More highlights should be provided to ClouDedup, such as retrievability and data integrity proof. The study[71] provided their proposal with the audit of the sentiment analysis concepts on Twitter, illuminating the techniques discovered and models employed along with a condensed Python-based methodology. A sack of words model, a form of content unigram model, is created using the NLTK toolkit. Python has been used for logical processing using the Scikit-learn library and NumPy basic module. It does, however, have application problems with the slang used, and the abbreviated forms of many of the phrase analyzers need to perform better as the number of classes is increased.

To ensure the secrecy and security of the information, Reference 72 proposed a straightforward information protection paradigm in which data is encrypted using the Advanced Encryption Standard (AES) before being sent into the cloud. Cloud computing is not an exception to the security and protection concerns that are true in the processing world. By shifting the encryption and decryption process from the cloud to the self, they provide the engineering and guidelines to scale up the security and protection of the information owner. This design may make it more difficult for a cloud provider to misuse or mine customer data. There should be tight restrictions on planning and computation to prevent the misuse of cloud computing's capabilities. When CSP can enable search on encrypted material, this is possible.[73]

Reference 74 proposed convergent encryptions as an easy way to make de-duplication compactable with encrypted data. De-duplicating encrypted data in the cloud while maintaining security is a difficult problem. They claimed it is difficult to use the cloud to cut down on duplication and compromise security. It can be improved even more with new storage optimization methods.

Reference 75 examined the various encryption techniques in 2016 and evaluated the performance of many symmetric algorithms. Among them are DES, 3DES, RSA, AES, and Blowfish. DES are block figures that encrypt and decrypt data using a common secret key. The DES technique starts with a string of a set length in plain text bits and transforms it into a string of ciphertext bits, with each block being 64 bits throughout many operations. The handling process consists of 16 similar stages called rounds. The initial permutation (IP) and final permutation (FP) are additional starting and finishing permutations (final permutation). 3DES is an improvement of DES, which has a key size of 192 bits and a 64-bit size (N. Vurukonda, B.T. Rao, and T Jiang, 2016). Like the first DES, the encryption approach involves increasing encryption and the average safe time. In CC, de-duplication emerges as an active research topic. This section comprises the research related to the work done in data de-duplication incorporating convergent encryption. The baseline approach and Dekey are the two distinct approaches recommended by Li et al.[16] for CE key management. Two significant deployment concerns plague the default strategy. First off, because it generates several keys, its efficiency is quite high. Because each user is required to safeguard their master key, it is also unreliable. The user also loses the data if the master key is lost.

Dekey, however, makes use of RSSS (Ramp Secret Sharing Scheme). The encryption key is distributed using this method to the various key servers. Instead of distributing the key, the author uses a key generation method in the cloud as a service to empower the key to authorized consumers. Youn et al.[76] suggest a different method for creating the key with access privileges for convergent encryption. Only individuals with the appropriate authorizations can access the shared data to create a convergent key. This method uses the DupLESS scheme's RSA blind signature based oblivious PRF protocol to generate a key from the key server.[77] Privileged information has been used in this technique to restrict access to the data to authorized users only. This system consists of users, cloud storage providers, and an authorization server, among other entities. AS is employed for the creation and administration of private keys. Additionally, based on a user's privileges, it is utilized to calculate a convergent encryption key for a particular file. The authors, however, had to have clarified how someone who has the same data copy might obtain the convergent key.

DICE (Dual Integrity Convergent Encryption), a secure data de-duplication technique, was proposed by Agarwala et al.[78] It is focused on (i) removing threats and preventing duplicate faking and (ii) providing integrity checks at both client and server ends. The produced tag is uploaded to the server, where it is subsequently subjected to an integrity check. Only when the client downloads the tag to access the ciphertext is this check run. As a result, bandwidth usage is decreased because only the tag is sent rather than a lengthy ciphertext, and de-duplication is also accomplished at the same time. The main generating and management method needs to be thoroughly defined in this article.

Nearly 80% of the firms surveyed by DuBois et al.[79] revealed that they were looking into data de-duplication technologies for their storage frameworks to reduce redundant data and thereby increase storage effectiveness and lower storage expenses. Many academics have already suggested many methods for data de-duplication in cloud storage with CE.

Guo et al.[80] provide a client-side de-duplication system (R-Dedup) that is randomized and secured. Both peer users and reliance on any third party are not prerequisites for this system. For users who have the same file copy, an encryption key is generated. In the data verification stage, R-Dedup also provides user authentication for the cloud server, ensuring data integrity. R-dedup provides a straightforward architecture with increased security.

A Secure De-duplication and Virtual Auditing of Data in the Cloud (SDVADC) mechanism is proposed by Geeta et al.,[81] which efficiently deduplicates the data of encoded information. Furthermore, a virtual Auditing Entity (VAE) is in-built into this proposed mechanism which virtually supports efficient auditing of the data owner's file during the download process.

Gao et al.[82] proposed a low-entropy secure data auditing scheme having file and authenticator de-duplication. The computation of authenticators and tagging of a file is designed in a new way. In this scheme, one copy of the data block and authenticators for the duplicated file is stored in the cloud.

Wang et al.[83] focus on the secure de-duplication scheme and efficient user revocation. Multi-user updatable encryption is proposed first, which helps the data owner to update the ciphertext efficiently for a new group of users. Then by using this technique, a new de-duplication scheme is constructed. While updating the data authority, a token is to be sent to the cloud by the data owner. Only then will the cloud update the ciphertext for a new group.

In this research, Zhang et al.[84] present a blockchain-based de-duplication technique for the cloud. To achieve the approved de-duplication, a novel hierarchical role hash tree (HRHT) is also created, which maps the relationship between the user's role and the role key.

5 | CONVERGENT ENCRYPTION FOR SECURE DATA DE-DUPLICATION

The combination of Convergent Encryption (CE) and Data Deduplication is crucial for maintaining the benefits of integrating both technologies while addressing the challenges faced by CE, such as "Dictionary attack," "Confirmation of File (CoF)," and "Learn-the-Remaining information (LRI)." ClouDedup, proposed by Puzio et al. in 2013, offers data confidentiality through CE throughout the deduplication process. Each original data copy is used to generate a Convergent Key (CK), which is then used to encrypt copies of the data by users or data owners. The user provides a tag that will be used to distinguish duplicate copies of the data for each copy of the data. The user sends the tag to the server first to check for duplication to see whether the same copy has already been stored. The CK and tag are both separately obtained, hence maintaining the confidentiality of data cannot be done by using the tag to discover the CK. The server side will store the encrypted data copy and its accompanying tag.

5.1 | CE Algorithm

The followings are the phases of the CE algorithm:

Steps for CE Algorithm
Step 1: $KeyGen_{CE}$ (M)
Step 2: Enc_{CE} (K, M)
Step 3: Dec_{CE} (K, C)
Step 4: TagGen (M)

Description of CE Algorithm
Step 1: KeyGenCE (M): In this step, the algorithm generates a secret key, CEK (Convergent Encryption Key), based on the client's data, CD. CD is converted to binary representation and divided into blocks. The blocks are processed to create the CEK.
Step 2: EncCE (K, M): Using the CEK (K) obtained from KeyGenCE, the algorithm encrypts the message, M. The encryption involves splitting the message into even and odd blocks, combining them to form a middle block, and converting it to binary representation.
Step 3: DecCE (K, C): Using the CEK (K) obtained from KeyGenCE, the algorithm decrypts the ciphertext, C. The decryption process involves reversing the encryption steps and obtaining the original message.
Step 4: TagGen (M): The algorithm generates a tag, or a unique identifier, for the message, M. This tag can be used for verification or identification purposes.

Overall, the CEKGA (Convergent Encryption Key Generation Algorithm) algorithm performs key generation, encryption, decryption, and tag generation using the client's data, CD.

A key generation algorithm K, CK is mapped to a copy of the data M. The inputs CK, K, and the data copy M are all accepted by the symmetric encryption technique C, which then returns a ciphertext. The ciphertext C and the CK, K inputs to the decryption mechanism M, which outputs the original data copy M. The tag-creation algorithm, T (M), translates the original data copy M to the tag T (M).

5.2 | Steps for CE by a user Alice

- Get a file F. Alice derives an encryption key K from the file by applying SHA-256 to file F.
- Then we have to encrypt that file F into ciphertext C with AES under key K.
- Protect the key K by encrypting it into W using its public key.
- Upload or send both C and W, ensuring that C and W are stored together.

5.3 | Alice can decrypt the encrypted file C into the original file F by

- Receiving the C and W files from the site/storage and downloading them.
- Using a personal decryption key, extract the key K from W.
- To recover the original file F, decrypt the ciphertext C using the AES algorithm and key K.

5.4 | CE using SHA-256 and AES

```
From convergent import CE
C1 = CE ("Secret Code")
Key, block id, ciphertext = C1•Encrypt ("ShahnawazAhmad")
If (len (ShahnawazAhmad) = = len (ciphertext))
True
C2 = CE ()
Plain_text = C2•Decrypt (key, ciphertext)
Plain1 text = = ShahnawazAhmad
true
```

The given code snippet demonstrates the use of Convergent Encryption (CE) with SHA-256 and AES algorithms.

In the first step, the string "Secret Code" is encrypted using CE, resulting in C1. Then, the encryption process continues with the encryption of the string "ShahnawazAhmad" using the key and block ID obtained from C1. The resulting ciphertext is stored in the variable "ciphertext." If the length of the original string "ShahnawazAhmad" is equal to the length of the ciphertext, it returns true. In the next step, a new instance of CE, represented by C2, is created. The "ciphertext" is then decrypted using the key and C2's decryption function, resulting in the plaintext. If the decrypted plaintext matches the original string "ShahnawazAhmad," it returns true. To summarize, SHA-256 and AES are utilized within the CE process to ensure secure encryption and decryption of the data.

Using the user's unique information, CE creates a key (hash value) and uses it to jumble the data. The identical information will be jumbled using this technique, which can help identify duplicate data. Additionally, a technique for creating the CK by processing the client's data was suggested by this inquiry. The CEKGA is denoted as $CE_K = CE_{keyGen}(C_k)$. CE_K is a CE key, and $CE_{keyGen}(C_k)$ is a primitive function for generating CEKGA. The unique phases in the proposed algorithm are followed by the suggested CEKGA, which is a digested algorithm that reads the client's input and produces the CE_K.

5.5 | Pseudo code of CEKGA

SCE_{keyGen} (C_k)

1. $C_D \leftarrow$ Client's data
2. Ns $\leftarrow$ size of (C_D)
3. C_D [S] $\leftarrow$ array (C_D)
4. for i $\leftarrow$ 1 to Ns
   $B_{SC}C_D$[i] $\leftarrow$ ASCII (C_D[i])
   next i
5. j $\leftarrow$ 1
6. k $\leftarrow$ 1
7. for i $\leftarrow$ 1 to Ns
   if (i%2= = 0) then
   E_{BLOCK} [k] $\leftarrow$ $B_{SC}C_D$[i]
   k $\leftarrow$ k + 1
   else
   O_{BLOCK} [j] $\leftarrow$ $B_{SC}C_D$[i]
   j $\leftarrow$ j + 1
   end if
   next i
8. Middle $\leftarrow$ Ns/2
9. for i $\leftarrow$ 1 to Middle
   S_{BLOCK} [i] $\leftarrow$ E_{BLOCK} [j] + O_{BLOCK} [j]
   next i
10. for i $\leftarrow$ 1 to Middle
    $BinU_D \leftarrow$ append (binary (S_{BLOCK} [i]))
    next i
11. BinNs $\leftarrow$ size of ($BinC_D$)
12. Blk $\leftarrow$ BinNs/256
13. n $\leftarrow$ 1
14. Bin $\leftarrow$ 256 0's block
15. for i $\leftarrow$ 1 to Blk
    Binck[i] $\leftarrow$ split ($BinC_D$, n, n + 255)
    n $\leftarrow$ n + 256
    Bin $\leftarrow$ 0's & 1's addition (Bin, Binck[i])
    next i
16. $BinC_D \leftarrow$ B
17. Blk $\leftarrow$ size of ($BinC_D$)/8
18. n $\leftarrow$ 1
19. for i $\leftarrow$ 1 to Blk
    $DecC_D$ [i] $\leftarrow$ deci (split (BinUp, n, n + 7))
    $BscC_D$ [i] $\leftarrow$ ASCII ($DecC_D$ [i])
    BscBuff $\leftarrow$ append ($BscC_D$[i])
    n $\leftarrow$ n + 8
    next i $\oplus$ C_k
20. $CE_K \leftarrow$ BscBuff
21. End S

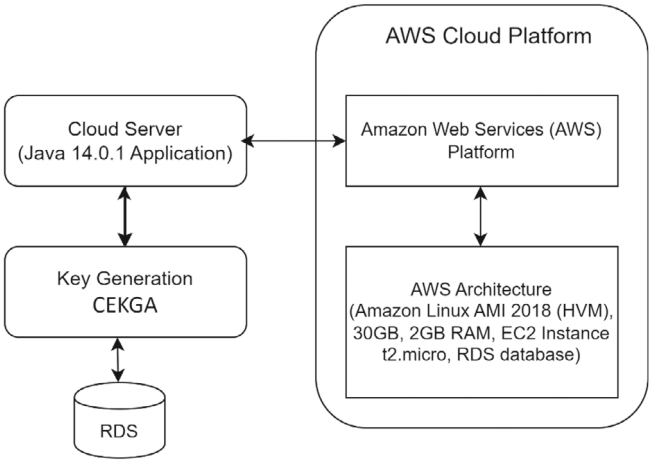


FIGURE 8 Block diagram of simulation cloud environment of CEKGA.

The given code, CEkeyGen, performs key generation for convergent encryption. It takes the client's data, CD, and converts it into an array of ASCII values, BSCCD. The array is then split into even and odd blocks, EBLOCK and OBLOCK respectively. A middle block, SBLOCK, is created by concatenating corresponding elements from EBLOCK and OBLOCK. SBLOCK is converted to binary, and the resulting binary strings are appended to form BinUD. BinUD is further processed by splitting it into blocks of 256 bits, Binck, and performing binary addition to obtain BinCD. BinCD is then split into groups of 8 bits, DecCD, which are converted back to ASCII and appended to BscBuff. Finally, Ck is XORed with BscBuff to obtain the convergent encryption key, CEK.

The user can utilize the generated CE_k to use encryption to encrypt data using the pseudo-code. The suggested technique is sometimes called a digest algorithm because it breaks down the user's input into key-based data content. The key generation and management function provides a key utilized to create the algorithm proposed by CE_k (KGMS). The key does not have to be carried about by the user. You can safely remove this key. CE_k is also saved in KGMS and is linked to Ck since the key is checked by KGMS and provided to users once they submit and prove their ownership of the data. Users must verify their data ownership to retrieve these keys from KGMS.

The following are the details of the environment used for the simulation. Amazon Linux AMI 2018.03.0 (HVM) miniaturized scale occurrence with 30 GB storage and 2 GB RAM, EC2 instance t2.micro, and RDS database are leased from AWS cloud foundation—a cloud server for cloud storage. The proposed CEKGA for the C_k era is developed using Java 14.0.1, and it is facilitated within the stage given by the AWS platform. A primary cloud service is provided by the cloud server, where this CEKGA is additionally included. All these handles are coded in the primary cloud service, which is sent within the AWS PaaS from AWS. The simulation of the cloud environment is represented diagrammatically in Figure 8.

5.6 | Proposed secure data de-duplication

The proposed algorithm is shown in Figure 9. Starting with the SHA-1 hash of the uploaded folder as the document level, Token Req (Tag, UserID) asks the private server (PS) for a token along with File Tag (File), and DupCheckReq (Token) asks the storage server (SS) to check the file for duplicates by sending the file token obtained from the private server. ShareTokenReq (Tag, "Priv.") requires the PS to create the shared file token with the target sharing privilege set and the File Tag (File). FileEncrypt (File) encrypts the file with CE using the 256-bit AES algorithm in Cipher Block Chaining (CBC) mode, and the convergent key is derived from the file's SHA-256 hash. FileUploadReq (FileID, File, Token) uploads the File Data to the SS if the file is distinct and the changes of the File Token (FT) are saved. The PS is implemented with matching request handlers for token creation and supports HashMap key storage. TokenGen (Tag, UserID) loads the user's associated opportunity keys and generates an HMAC-SHA-1 token. Based on the ideas of MD4 and MD5, SHA-1 generates an MD. Only one bitwise rotation in the message scheduling of its compression mechanism distinguishes SHA-1 from SHA-0. MD is a 160-bit hash value generated using SHA-1. This hash value is represented as a 40-digit hexadecimal number.

The suggested methodology is employed at a medical facility so that a doctor can access it remotely. De-duplication is used at a healthcare facility to expand its cloud storage capacity. More storage space is available when data duplication is avoided. In medical facilities, the data will be kept in files with various formats and records, and its report will be frequently updated based on the check-ups. Due to the bulk of medical pictures like MRI, ultrasound scans, and echocardiography reports, data duplication lowers storage capacity in such a case.

Additionally, the security element of this medical data is crucial because any security breach could result in the disclosure of the patient's personal information. Therefore, a data de-duplication method and convergent encryption method are suggested as solutions to this issue to increase

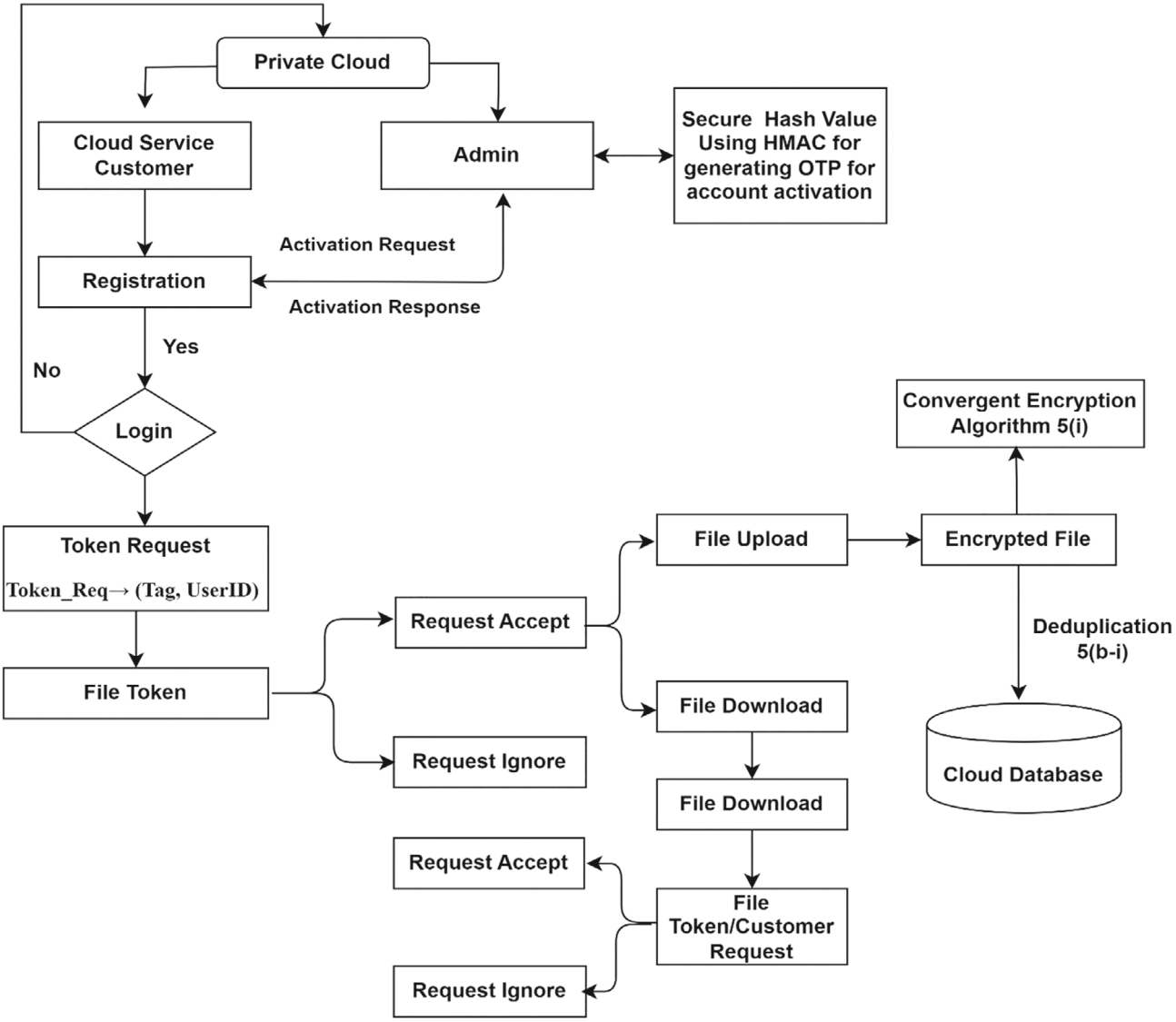


FIGURE 9 Block diagram of the proposed framework.

information confidentiality and storage capacity. First, redundant data is eliminated to conserve storage space in the cloud. Then, the various file types are subjected to the data de-duplication procedure. The following describes how the suggested methodology operates:

- After the user completes the login process, the upload question will show up. By choosing "yes," the user can upload files containing the data. The upload will stop if the user chooses "No."
- After selecting the image, the cloud administrator will search for duplicate images. If there are no duplicate photos, the image will be posted immediately; otherwise, the ownership will be checked to ensure a successful upload. The original image kept on the local server will be used instead if the ownership of the image cannot be confirmed. The following procedure is used to check for image duplication and the uploading process:
    - File Encryption.
    - Generation of Hash key.
    - Searching for duplicates.
    - File shared with the user.

5.6.1 | Pseudo code of the de-duplication

1. File_Tag ← File
2. Token_Req ← (Tag, UserID)
3. DupCheckReq ← Token
4. ShareTokenReq ← (Tag, {Priv.})
5. FileEncrypt ← File
6. FileUploadReq ← (FileID, File, Token)
7. TokenGen ← (Tag, UserID)

The provided code consists of a series of steps involving file tagging, token generation, duplicate checking, sharing tokens, file encryption, and file upload requests. Initially, a file tag is assigned to a file. Then, a token request is made, which includes the file tag and the user ID. A duplicate check request is sent using the token obtained from the previous step. Following that, a share token request is made, containing the file tag and a set of privileges. The file is encrypted using a file encryption process. Next, a file upload request is sent, which includes the file ID, the encrypted file, and the token. Finally, a token generation process is initiated, involving the file tag and the user ID. Overall, these steps illustrate a flow involving tagging, tokenization, duplicate checking, encryption, and file upload in a file management system or similar context.

5.6.2 | System model of proposed scheme

Figure 10 shows the proposed system model of the cloud environment. It contains Client (C), key generation, and Cloud Storage (CS), where the data de-duplication with CE occurs. The process is explained in the given steps below:

- First, a token Tkn is generated from the uploaded client data (C_D) using $T_{KN} = Tkn_Gen(C_D)$ primitive function.
- The token T_{kn} is then forwarded to the key generation to get a key for generating CEk through a secure channel.
- KeyGen has then verified the metadata for the existence of Tkn in the database. In case of the presence of T_{KN} in the database, KeyGen will forward the key to the user corresponding to the same Tkn. In case of its non-existence in the database, a key (GCK) is generated and sent to the user by KeyGen.
- The user then uses this forwarded key GCK by KGMaaS for generating convergent encryption key CEK.

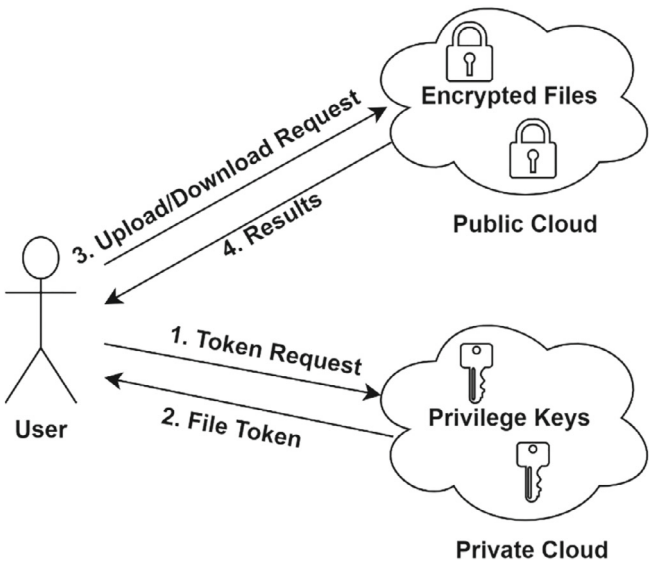


FIGURE 10 Block diagram of de-duplication based on the cloud environment.

TABLE 5 Data classification in the proposed framework.

Classification	Basic capacities	Description
Data Classification in the proposed framework[64]	KeyGenSE (1)! *K*	This is an algorithm used to develop κ through parameter 1.
	EncSE (κ, *M*)! *C*	This algorithm is responsible for hiding the secret of κ and *M*, respectively. It also outputs the coded text *C*.
	DecSE (κ, *C*)! *M*	Unlike the other algorithms, DecSE (κ, *C*)! *M* is used for decryption. It is used to take the secret κ and coded text *C*.
	KeyGenCE (M)! K	This is a key generation algorithm. It assists in illustrating data copy M that is convergent to K.
	EncCE (K, M)! C	This is the symmetric encryption system.
	DecCE (K, C)! M	This is the decryption algorithm.
	TagGen (M)! T (M)	This is the tag generation algorithm representing the original data copy M.

- Clients generate the CEk using the proposed algorithm $CE_K = CE_{keyGen}(C_k)$ with the key GCK received from the KeyGen.

Table 5 shows the data classification based on primary storage and description.

The various UML diagrams, including the class diagram, use case diagram, activity diagram, and sequence diagram, are shown in Figures 11–14, have been used to demonstrate the effectiveness of the suggested scheme.

5.7 | Experiments and analysis of the proposed approach

The experimental results and their analysis are reported in this section. Java was used to accomplish the suggested plan on Amazon EC2 servers with Intel i7 preprocessors. Using data frequently received on mobile devices, we chose eight different datasets (generated by the authors), that is, DS 1, DS 2, DS 3, … , DS 6. DS 1 consists of text files (.text). DS 2 consists of Java application data (.java), DS 3 consists of CSS application data (.css), DS 4 consists of HTML application data (.html), DS 5 consists of JavaScript Application data (.jss), DS 6 consists of React application data (.jsx). The size of data blocks before de-duplication was taken to be 2.08, 1.80, 1.55, 2.50, 2.66, and 1.66 kB (as shown by Table 6 and Figure 15). By applying the proposed approach of data de-duplication on these DS, the size of data blocks becomes 1.90, 1.60, 1.30, 2, 2.20, and 1.20 kB (as shown in Table 6 and Figure 16). The files can be broken into size blocks to make share management easier; however, this will reduce the file's de-duplication rate.

The experiment evaluated the proposed de-duplication approach using eight datasets (DS1–DS8) representing various file types commonly found on mobile devices (text, Java code, CSS, HTML, JavaScript, and React). The analysis focuses on two key aspects: de-duplication effectiveness and impact on file size. The experiment demonstrates that the proposed approach achieves de-duplication across all datasets. This is evident from the reduction in file sizes reported in Table 7 and Figure 16 compared to Table 6 and Figure 15. The reduction percentages range from 9% (DS1) to 29% (DS6), indicating that the scheme can identify and eliminate redundant data blocks within and potentially across datasets. Here's a breakdown of the de-duplication effectiveness:

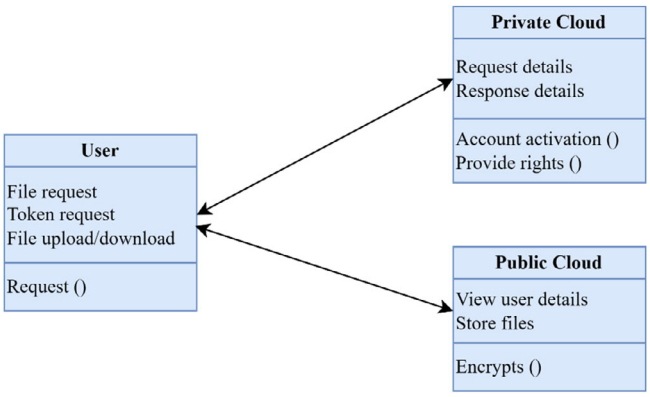


FIGURE 11 Class diagram of cloud environment.

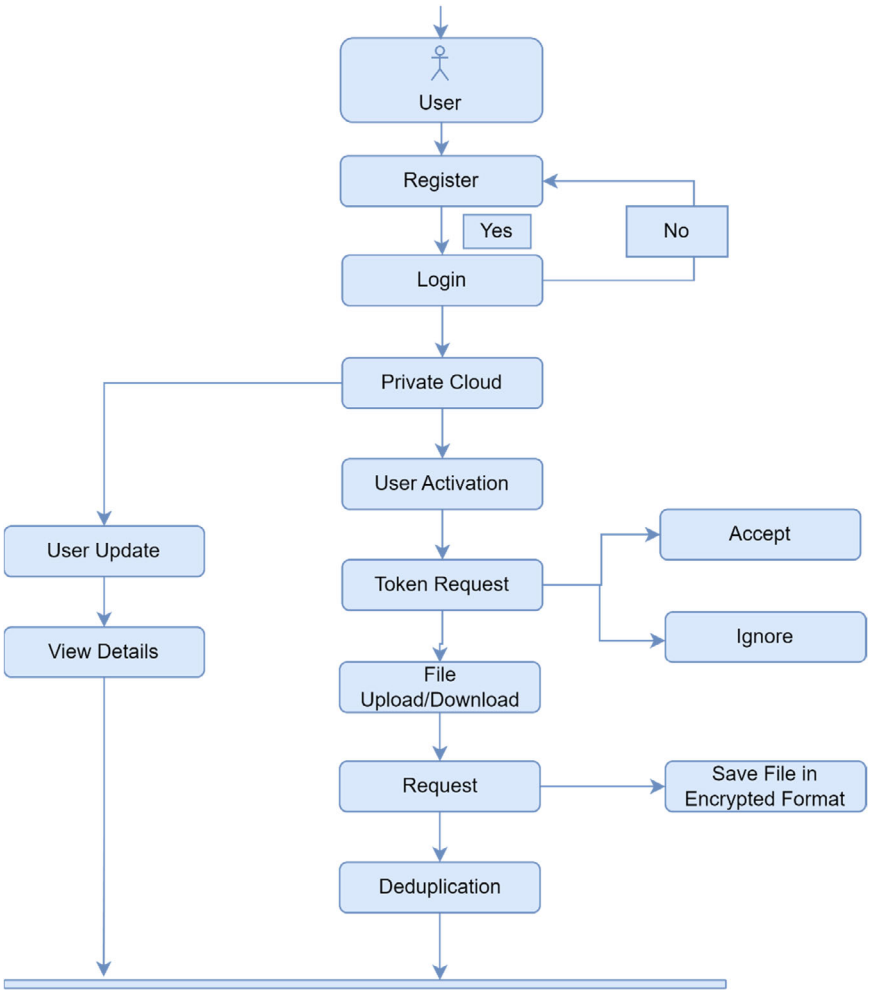


FIGURE 12 Activity diagram of data de-duplication.

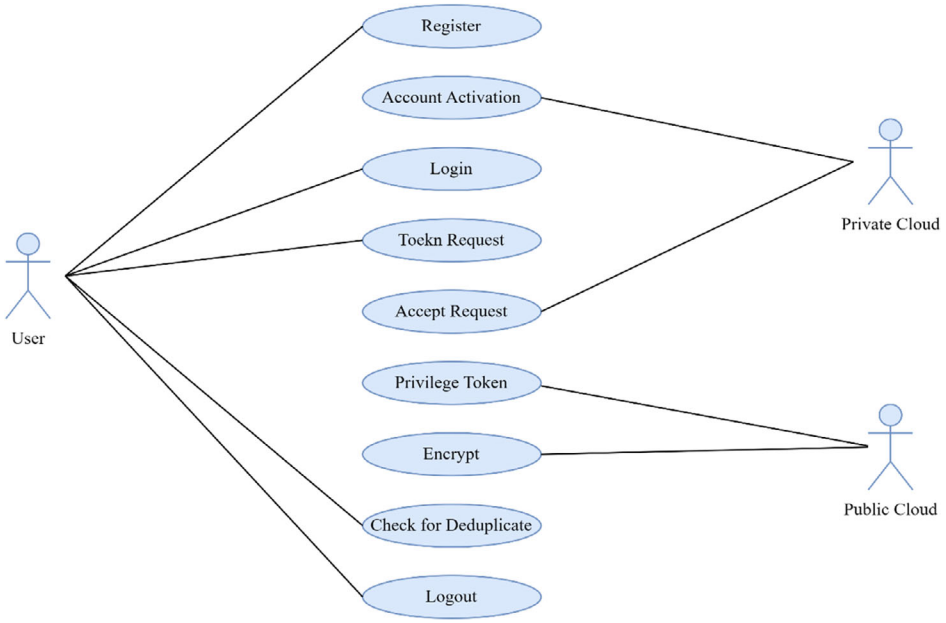


FIGURE 13 Use case diagram of data de-duplication.

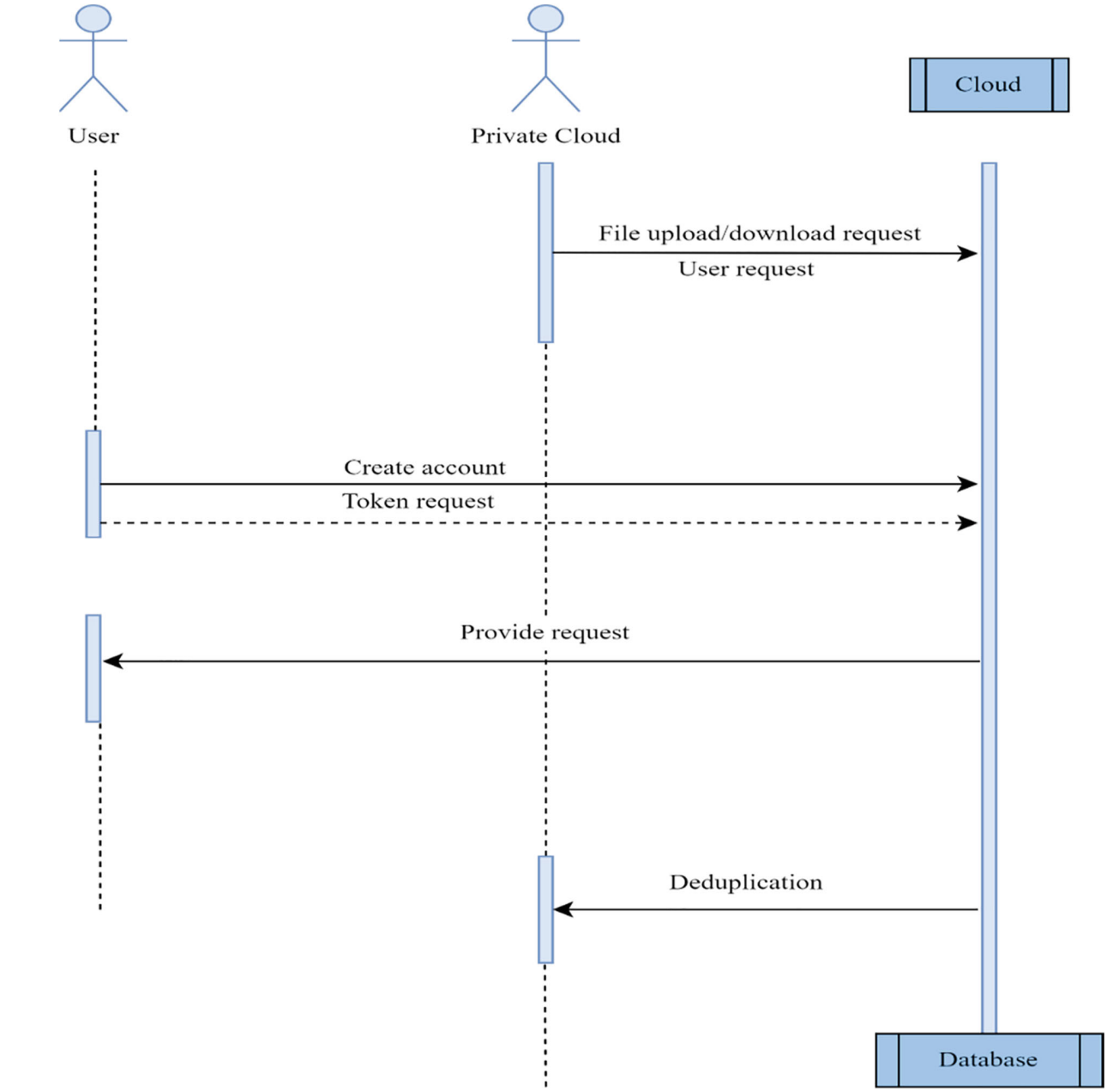


FIGURE 14 Sequence diagram of data de-duplication.

TABLE 6 Size of files before de-duplication.

Data sets	File name	File extension	File size in kB
DS 1	data	.text	2.08
DS 2	abc	.java	1.80
DS 3	boot	.css	1.55
DS 4	abc	.html	2.55
DS 5	script	.js	2.66
DS 6	script	.jsx	1.66

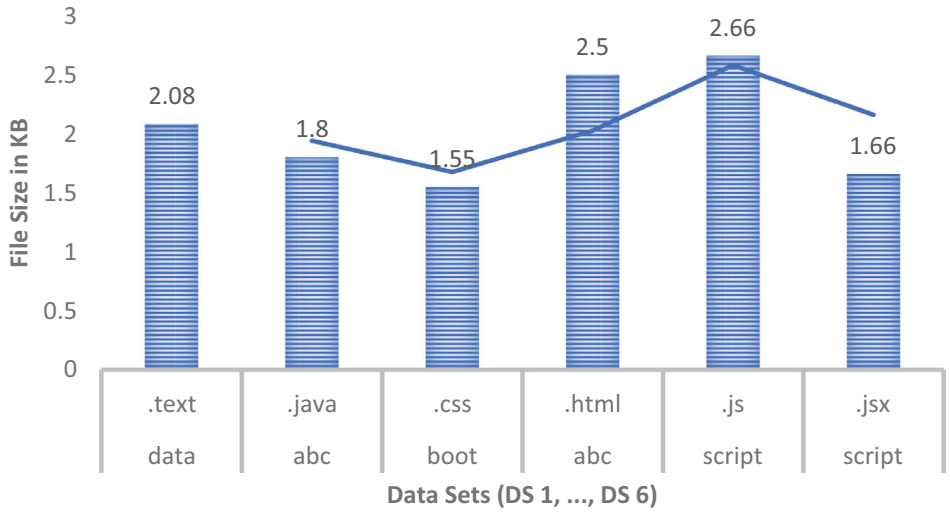


FIGURE 15 Size of files before de-duplication.

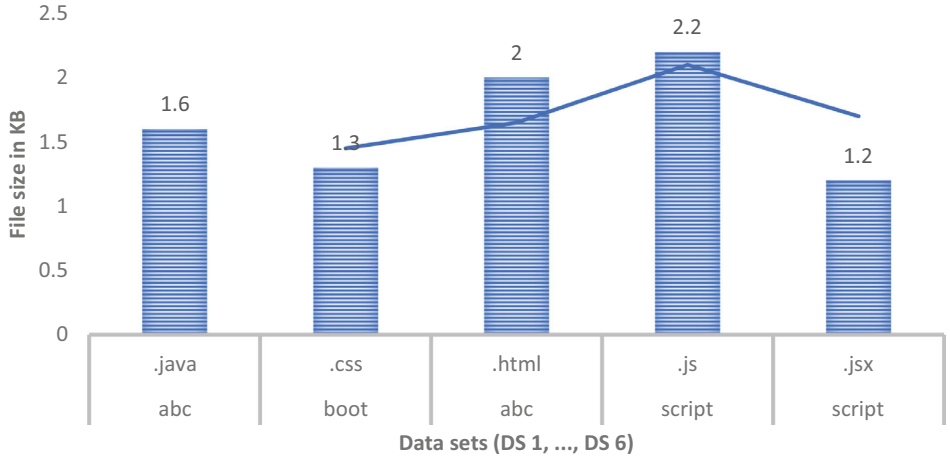


FIGURE 16 Size of files after de-duplication.

TABLE 7 Size of files after de-duplication.

Data sets	File name	File extension	File size in kB	Saved space in kB
DS 1	data	.text	1.90	0.18
DS 2	abc	.java	1.60	0.20
DS 3	boot	.css	1.30	0.25
DS 4	abc	.html	2.00	0.55
DS 5	script	.js	2.20	0.46
DS 6	script	.jsx	1.20	0.46

TABLE 8 Comparative security analysis of the proposed scheme.

De-duplication issues	Bellare et al.[77]	Halevi et al.[85]	Koo and Hur[86]	Ng et al.[87]	Xu et al.[53]	Proposed scheme
Encrypted data de-duplication	√	x	√	x	√	√
Tags for preserving consistency	√	x	√	x	√	√
Updating data that has been outsourced	x	x	√	x	x	√
De-duplication on the client side	√	x	√	x	√	√
Repeated heterogenous files	x	x	x	x	x	√

- *Higher reduction for larger files*: Datasets with larger initial sizes (DS4, DS5) generally achieved a greater percentage reduction after de-duplication (22% and 19%, respectively). This suggests the proposed approach is more effective in identifying redundancy in larger data blocks.
- *Data type variations*: The de-duplication effectiveness seems to vary depending on the data type. DS3 (.css) achieved a higher reduction (18%) compared to DS1 (.text) with a similar initial size (around 2 kB). This might be due to inherent redundancies present in CSS code structure.

The de-duplication process successfully reduced the size of all data blocks. The "Saved Space" column in Table 6 quantifies this reduction in kilobytes (KB). This space saving translates to benefits like:

- *Reduced storage costs*: Smaller file sizes lead to lower cloud storage requirements, potentially reducing storage costs associated with mobile device data backups.
- *Improved transmission efficiency*: Smaller data packets can be transmitted faster, especially over bandwidth-constrained mobile networks.

Table 7 shows the outcomes of de-duplication when different dataset sizes were employed. The size of DS 1 was reduced by around 9%, the size of DS 2 was reduced by about 13%, the size of DS 3 was reduced by about 18%, the size of DS 4 was reduced by about 22%, the size of DS 5 was reduced by about 19%, and the size of DS 6 was reduced by about 29%.

We can save cloud storage space using a de-duplication strategy on these files.

5.8 | Security analysis of the proposed scheme

This section examines the security of the proposed scheme before comparing its functionality and effectiveness to other related works. The suggested scheme is compared to existing schemes in Table 8 based on the de-duplication difficulties the system provides.

Existing encryption solutions, such as traditional symmetric (AES) or asymmetric encryption, often face limitations regarding deduplication due to their reliance on fixed keys, which do not utilize the data itself for encryption, making duplicate identification challenging. On the other hand, existing deduplication solutions like Content-addressed Storage (CAS) employ unique identifiers (hashes) to identify and store data blocks but may compromise confidentiality as the data content can be derived from the hash. The proposed system aims to address several challenges, including vulnerability to dictionary attacks in traditional encryption methods, which could be mitigated by the Convergent Encryption (CE) approach that derives keys from the data, potentially offering protection against such attacks. Additionally, confirming data integrity post-deduplication, known as Confirmation of File (CoF), poses challenges in some existing solutions, but the proposed system's use of tags may facilitate CoF by enabling verification of data content post-deduplication. Moreover, some deduplication schemes are susceptible to Learn-the-Remaining Information (LRI) attacks, where attackers can infer information about the original data from encrypted data blocks. The proposed CE, employing data-dependent keys, may provide defense against LRI attacks.

6 | CONCLUSION AND FUTURE RESEARCH DIRECTION

This paper presented a novel approach to secure data deduplication in the cloud. The proposed system leverages convergent encryption (CE) and a Convergent Key Generation Algorithm (CEKGA) to enable secure deduplication on encrypted data. CEKGA simplifies key management by generating encryption keys based on user data. This approach offers several strengths, including enhanced data security through encryption, improved storage efficiency by eliminating redundancy, and reduced user burden for key management. However, a detailed security analysis and performance evaluation are necessary to fully understand the system's robustness and practicality. Additionally, comparing this CE-based deduplication with existing techniques would further solidify its contribution. Future work can involve implementing the system on a real cloud platform, conducting

a thorough security analysis, and exploring its applicability to various data types beyond text and multimedia. By addressing these limitations and pursuing further research, this approach has the potential to become a significant advancement in secure and efficient cloud storage solutions.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no competing interests.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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